Course 5043C | **Semester V** | **Theory** | **4 Credits**

Biomedical Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5043C as Biomedical Engineering in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Biomedical Instrumentation course and polytechnic biomedical curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Medical device calibrations, clinical diagnoses, safety settings and hospital plans must be based on approved clinical protocols, certified medical hardware data, current healthcare standards and competent professional review.

Verified source note: Verified sources support bio-potentials, sensors, therapeutic equipment (pacemaker, ventilator), and diagnostic imaging. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Biomedical Engineering studies the application of electronics and engineering principles to the field of medicine and biology. It develops the ability to analyze bio-potentials (ECG, EEG, EMG), evaluate bio-electrodes and sensors, understand therapeutic and diagnostic equipment (Pacemakers, X-ray, MRI), and coordinate patient safety standards while respecting the technical and clinical boundaries of medical instrumentation.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electronics, human anatomy and physiology basics, transducers and sensors, and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the origin and characteristics of physiological signals and bio-potentials in the human body.
2. Explain the working principles and selection of bio-electrodes and sensors for medical measurements.
3. Explain the operation and maintenance of therapeutic equipment like defibrillators, pacemakers and ventilators.
4. Explain the principles of diagnostic imaging (X-ray, CT, MRI) and the technical requirements for patient monitoring and hospital safety.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain the origin of bio-potentials and the characteristics of physiological signal acquisition.	Understand
CO2	Explain the principles of bio-electrodes and transducers for medical instrumentation.	Understand
CO3	Explain the operation of cardiovascular, respiratory and therapeutic medical equipment.	Understand
CO4	Explain diagnostic imaging techniques and hospital safety standards for medical electronics.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Bio-potentials and Physiological Signals	Cell structure and resting/action potentials; origin of bio-potentials; ECG (Electrocardiogram); EEG (Electroencephalogram); EMG (Electromyogram); physiological systems overview (Cardiovascular, Nervous, Respiratory).	Approx. 14 hours	CO1	Module 1
2	Bio-electrodes and Sensors	Electrode-electrolyte interface; polarization and non-polarizable electrodes; surface electrodes (Ag/AgCl); internal electrodes (microelectrodes); transducers for physiological measurements (temperature, pressure, flow).	Approx. 14 hours	CO2	Module 2
3	Therapeutic and Cardiovascular Equipment	ECG machine block diagram; cardiac pacemakers (Internal and External); Defibrillators (AC and DC); Heart-lung machine; Ventilators; Hemodialysis machine; Cardiac output measurement.	Approx. 16 hours	CO3	Module 3
4	Diagnostic Imaging and Patient Safety	X-ray machine (generation and detection); Computed Tomography (CT); Magnetic Resonance Imaging (MRI); Ultrasound imaging; Patient monitoring systems; Electrical safety: Micro-shock, Macro-shock, Grounding, Isolation.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Bio-potentials and Physiological Signals

Cell structure and resting/action potentials; origin of bio-potentials; ECG (Electrocardiogram); EEG (Electroencephalogram); EMG (Electromyogram); physiological systems overview (Cardiovascular, Nervous, Respiratory).

CO1 · Approx. 14 hours

Bio-electrodes and Sensors

Electrode-electrolyte interface; polarization and non-polarizable electrodes; surface electrodes (Ag/AgCl); internal electrodes (microelectrodes); transducers for physiological measurements (temperature, pressure, flow).

CO2 · Approx. 14 hours

Therapeutic and Cardiovascular Equipment

ECG machine block diagram; cardiac pacemakers (Internal and External); Defibrillators (AC and DC); Heart-lung machine; Ventilators; Hemodialysis machine; Cardiac output measurement.

CO3 · Approx. 16 hours

Diagnostic Imaging and Patient Safety

X-ray machine (generation and detection); Computed Tomography (CT); Magnetic Resonance Imaging (MRI); Ultrasound imaging; Patient monitoring systems; Electrical safety: Micro-shock, Macro-shock, Grounding, Isolation.

CO4 · Approx. 16 hours

MODULE 1

Bio-potentials and Physiological Signals

Cell structure and resting/action potentials; origin of bio-potentials; ECG (Electrocardiogram); EEG (Electroencephalogram); EMG (Electromyogram); physiological systems overview (Cardiovascular, Nervous, Respiratory).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

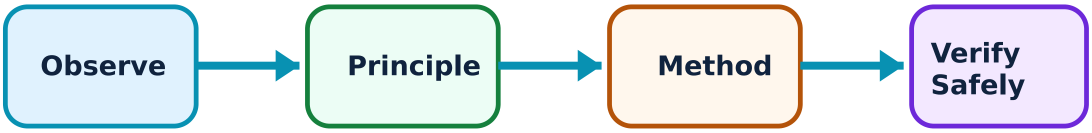
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bio-potentials and Physiological Signals: identify the system, state its principle, follow the correct method and verify the result safely.

1. Origin of bio-potentials−

Bio-potentials are generated by ion movement across cell membranes. The resting potential (~ -70mV) and action potential are fundamental. ECG records heart activity, EEG brain waves and EMG muscle activity. Each signal has specific amplitude and frequency ranges critical for accurate acquisition and diagnosis.

Study and application method

Describe the process of 'Depolarization' and 'Repolarization' in a cell; list the typical frequency and amplitude ranges for ECG and EEG signals.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the process of 'Depolarization' and 'Repolarization' in a cell; list the typical frequency and amplitude ranges for ECG and EEG signals.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Biomedical signals are extremely weak (microvolts to millivolts) and prone to noise. Do not perform signal acquisition without authorised skin preparation and high-quality shielded cables.

2. Physiological systems−

The cardiovascular system pumps blood, the nervous system controls functions through electrical impulses, and the respiratory system manages gas exchange. Understanding these systems is essential for designing and maintaining medical electronics that monitor their performance.

Study and application method

Sketch a typical ECG waveform and label the P, QRS and T waves; explain the significance of the 'QRS complex' in cardiac monitoring.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

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Common mistake / precaution: Medical signals can indicate life-threatening conditions. Do not interpret physiological waveforms for clinical diagnosis without authorised medical training and validated diagnostic software.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Bio-electrodes and Sensors

Electrode-electrolyte interface; polarization and non-polarizable electrodes; surface electrodes (Ag/AgCl); internal electrodes (microelectrodes); transducers for physiological measurements (temperature, pressure, flow).

Approx. 14 hours | CO2 | Understand · Apply

Learning goals

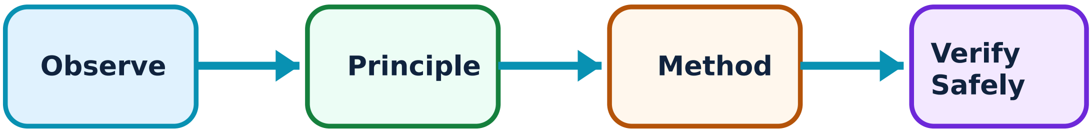
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bio-electrodes and Sensors: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bio-electrodes−

Electrodes convert ionic current in the body to electronic current in the instrument. The Ag/AgCl electrode is a standard non-polarizable electrode. Proper contact impedance is crucial for signal quality. Microelectrodes are used for intracellular recordings.

Study and application method

Explain the 'Electrode-Electrolyte Interface' conceptually; compare 'Surface Electrodes' and 'Needle Electrodes' in a conceptual table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Electrode-Electrolyte Interface' conceptually; compare 'Surface Electrodes' and 'Needle Electrodes' in a conceptual table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrodes must be biocompatible and sterile. Do not use electrodes for internal measurements without authorised sterilization and biocompatibility certification.

2. Medical transducers−

Transducers convert physiological parameters like blood pressure, body temperature and respiratory flow into electrical signals. Thermistors are used for temperature, while piezoelectric or strain-gauge sensors are used for pressure measurement.

Study and application method

Describe the working of a piezoelectric transducer for physiological pressure measurement; list three types of sensors used in patient monitoring.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of a piezoelectric transducer for physiological pressure measurement; list three types of sensors used in patient monitoring.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Transducer calibration is critical for patient safety. Do not use medical sensors without authorised periodic calibration and verification against reference standards.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Therapeutic and Cardiovascular Equipment

ECG machine block diagram; cardiac pacemakers (Internal and External); Defibrillators (AC and DC); Heart-lung machine; Ventilators; Hemodialysis machine; Cardiac output measurement.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

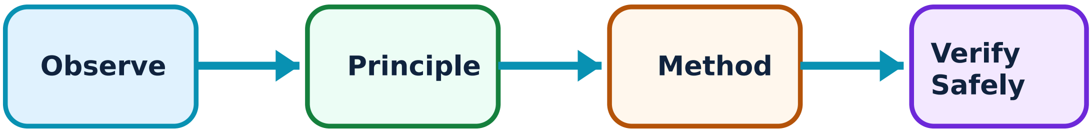
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Therapeutic and Cardiovascular Equipment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Cardiac therapeutic devices−

Pacemakers provide electrical pulses to maintain heart rhythm. Defibrillators deliver high-energy shocks to restore normal rhythm during cardiac arrest. These devices must be highly reliable and feature advanced safety interlocks to prevent accidental shocks.

Study and application method

Sketch the block diagram of a DC Defibrillator; explain the difference between 'Demand' and 'Fixed-rate' pacemakers.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of a DC Defibrillator; explain the difference between 'Demand' and 'Fixed-rate' pacemakers.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Therapeutic devices deliver energy to the patient. Do not operate defibrillators or pacemakers without authorised clinical training and strict adherence to safety protocols.

2. Respiratory and renal support−

Ventilators assist or replace spontaneous breathing. Hemodialysis machines filter waste from the blood when kidneys fail. These systems integrate complex fluidics, sensors and control logic to maintain life-critical physiological balances.

Study and application method

Describe the basic function of a 'Hemodialysis Machine'; list the key parameters monitored in a modern medical ventilator.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the basic function of a 'Hemodialysis Machine'; list the key parameters monitored in a modern medical ventilator.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Support equipment failure can be fatal. Ensure all life-support systems have authorised redundant power supplies, gas backups and validated alarm systems.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Diagnostic Imaging and Patient Safety

X-ray machine (generation and detection); Computed Tomography (CT); Magnetic Resonance Imaging (MRI); Ultrasound imaging; Patient monitoring systems; Electrical safety: Micro-shock, Macro-shock, Grounding, Isolation.

Approx. 16 hours | CO4 | Understand · Apply

Learning goals

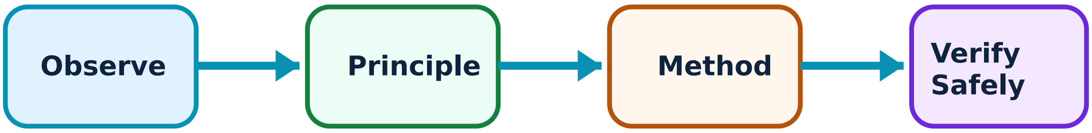
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Diagnostic Imaging and Patient Safety: identify the system, state its principle, follow the correct method and verify the result safely.

1. Diagnostic imaging techniques−

X-rays use ionizing radiation for bone and tissue imaging. CT provides cross-sectional views. MRI uses strong magnetic fields and RF pulses for high-contrast soft tissue imaging. Ultrasound uses high-frequency sound waves, providing a safe and real-time imaging modality.

Study and application method

Compare X-ray, CT and MRI in a conceptual table showing their imaging principles; describe the role of the 'Transducer' in ultrasound imaging.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare X-ray, CT and MRI in a conceptual table showing their imaging principles; describe the role of the 'Transducer' in ultrasound imaging.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: X-rays involve radiation hazards; MRI involves strong magnetic fields. Do not enter an MRI room with metallic objects; ensure all imaging areas follow authorised radiation safety and magnetic-field protocols.

2. Hospital safety and monitoring−

Patient safety is paramount. Macro-shock (milliamps) and micro-shock (microamps) are hazards. Isolation transformers, proper grounding and leakage current testing are essential. Bedside monitors track vital signs, providing alerts to healthcare staff.

Study and application method

Define 'Micro-shock' and explain why it is dangerous for catheterized patients; list three electrical safety measures used in hospital ICUs.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Micro-shock' and explain why it is dangerous for catheterized patients; list three electrical safety measures used in hospital ICUs.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrical leakage in medical environments can cause cardiac arrest. Do not connect medical equipment to non-hospital-grade outlets; ensure all devices undergo authorised periodic safety testing (e.g., IEC 60601).

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Bio-potentials and Physiological Signals

Use the concept flow to revise observation, principle, method and verification.

Module 2: Bio-electrodes and Sensors

Use the concept flow to revise observation, principle, method and verification.

Module 3: Therapeutic and Cardiovascular Equipment

Use the concept flow to revise observation, principle, method and verification.

Module 4: Diagnostic Imaging and Patient Safety

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare ECG, EEG and EMG in a conceptual table showing their signal characteristics and applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the block diagram of a generalized medical instrumentation system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the structure of a surface electrode and label the lead wire and electrolyte gel.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the therapeutic and diagnostic equipment found in a typical hospital emergency room or ICU.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

List the safety precautions required when working with X-ray or MRI equipment.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Bio-potentials and Physiological Signals

Explain Origin of bio-potentials and state one correct method, application or precaution.−

Bio-potentials are generated by ion movement across cell membranes. The resting potential (~ -70mV) and action potential are fundamental. ECG records heart activity, EEG brain waves and EMG muscle activity. Each signal has specific amplitude and frequency ranges critical for accurate acquisition and diagnosis.

Answer framework: Describe the process of 'Depolarization' and 'Repolarization' in a cell; list the typical frequency and amplitude ranges for ECG and EEG signals.

Important point: Biomedical signals are extremely weak (microvolts to millivolts) and prone to noise. Do not perform signal acquisition without authorised skin preparation and high-quality shielded cables.

Explain Physiological systems and state one correct method, application or precaution.−

The cardiovascular system pumps blood, the nervous system controls functions through electrical impulses, and the respiratory system manages gas exchange. Understanding these systems is essential for designing and maintaining medical electronics that monitor their performance.

Answer framework: Sketch a typical ECG waveform and label the P, QRS and T waves; explain the significance of the 'QRS complex' in cardiac monitoring.

Important point: Medical signals can indicate life-threatening conditions. Do not interpret physiological waveforms for clinical diagnosis without authorised medical training and validated diagnostic software.

Module 2 — Bio-electrodes and Sensors

Explain Bio-electrodes and state one correct method, application or precaution.−

Electrodes convert ionic current in the body to electronic current in the instrument. The Ag/AgCl electrode is a standard non-polarizable electrode. Proper contact impedance is crucial for signal quality. Microelectrodes are used for intracellular recordings.

Answer framework: Explain the 'Electrode-Electrolyte Interface' conceptually; compare 'Surface Electrodes' and 'Needle Electrodes' in a conceptual table.

Important point: Electrodes must be biocompatible and sterile. Do not use electrodes for internal measurements without authorised sterilization and biocompatibility certification.

Explain Medical transducers and state one correct method, application or precaution.−

Transducers convert physiological parameters like blood pressure, body temperature and respiratory flow into electrical signals. Thermistors are used for temperature, while piezoelectric or strain-gauge sensors are used for pressure measurement.

Answer framework: Describe the working of a piezoelectric transducer for physiological pressure measurement; list three types of sensors used in patient monitoring.

Important point: Transducer calibration is critical for patient safety. Do not use medical sensors without authorised periodic calibration and verification against reference standards.

Module 3 — Therapeutic and Cardiovascular Equipment

Explain Cardiac therapeutic devices and state one correct method, application or precaution.−

Pacemakers provide electrical pulses to maintain heart rhythm. Defibrillators deliver high-energy shocks to restore normal rhythm during cardiac arrest. These devices must be highly reliable and feature advanced safety interlocks to prevent accidental shocks.

Answer framework: Sketch the block diagram of a DC Defibrillator; explain the difference between 'Demand' and 'Fixed-rate' pacemakers.

Important point: Therapeutic devices deliver energy to the patient. Do not operate defibrillators or pacemakers without authorised clinical training and strict adherence to safety protocols.

Explain Respiratory and renal support and state one correct method, application or precaution.−

Ventilators assist or replace spontaneous breathing. Hemodialysis machines filter waste from the blood when kidneys fail. These systems integrate complex fluidics, sensors and control logic to maintain life-critical physiological balances.

Answer framework: Describe the basic function of a 'Hemodialysis Machine'; list the key parameters monitored in a modern medical ventilator.

Important point: Support equipment failure can be fatal. Ensure all life-support systems have authorised redundant power supplies, gas backups and validated alarm systems.

Module 4 — Diagnostic Imaging and Patient Safety

Explain Diagnostic imaging techniques and state one correct method, application or precaution.−

X-rays use ionizing radiation for bone and tissue imaging. CT provides cross-sectional views. MRI uses strong magnetic fields and RF pulses for high-contrast soft tissue imaging. Ultrasound uses high-frequency sound waves, providing a safe and real-time imaging modality.

Answer framework: Compare X-ray, CT and MRI in a conceptual table showing their imaging principles; describe the role of the 'Transducer' in ultrasound imaging.

Important point: X-rays involve radiation hazards; MRI involves strong magnetic fields. Do not enter an MRI room with metallic objects; ensure all imaging areas follow authorised radiation safety and magnetic-field protocols.

Explain Hospital safety and monitoring and state one correct method, application or precaution.−

Patient safety is paramount. Macro-shock (milliamps) and micro-shock (microamps) are hazards. Isolation transformers, proper grounding and leakage current testing are essential. Bedside monitors track vital signs, providing alerts to healthcare staff.

Answer framework: Define 'Micro-shock' and explain why it is dangerous for catheterized patients; list three electrical safety measures used in hospital ICUs.

Important point: Electrical leakage in medical environments can cause cardiac arrest. Do not connect medical equipment to non-hospital-grade outlets; ensure all devices undergo authorised periodic safety testing (e.g., IEC 60601).

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Bio-potentials and Physiological Signals.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Bio-electrodes and Sensors.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Therapeutic and Cardiovascular Equipment.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Diagnostic Imaging and Patient Safety.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Cell structure and resting/action potentials; origin of bio-potentials; ECG (Electrocardiogram); EEG (Electroencephalogram); EMG (Electromyogram); physiological systems overview (Cardiovascular, Nervous, Respiratory)..
2. Develop a complete answer or practical plan for: Electrode-electrolyte interface; polarization and non-polarizable electrodes; surface electrodes (Ag/AgCl); internal electrodes (microelectrodes); transducers for physiological measurements (temperature, pressure, flow)..
3. Develop a complete answer or practical plan for: ECG machine block diagram; cardiac pacemakers (Internal and External); Defibrillators (AC and DC); Heart-lung machine; Ventilators; Hemodialysis machine; Cardiac output measurement..
4. Develop a complete answer or practical plan for: X-ray machine (generation and detection); Computed Tomography (CT); Magnetic Resonance Imaging (MRI); Ultrasound imaging; Patient monitoring systems; Electrical safety: Micro-shock, Macro-shock, Grounding, Isolation..

LAST-MILE REVISION

Quick Revision

Module 1: Bio-potentials and Physiological Signals

Cell structure and resting/action potentials; origin of bio-potentials; ECG (Electrocardiogram); EEG (Electroencephalogram); EMG (Electromyogram); physiological systems overview (Cardiovascular, Nervous, Respiratory).

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Bio-electrodes and Sensors

Electrode-electrolyte interface; polarization and non-polarizable electrodes; surface electrodes (Ag/AgCl); internal electrodes (microelectrodes); transducers for physiological measurements (temperature, pressure, flow).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Therapeutic and Cardiovascular Equipment

ECG machine block diagram; cardiac pacemakers (Internal and External); Defibrillators (AC and DC); Heart-lung machine; Ventilators; Hemodialysis machine; Cardiac output measurement.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Diagnostic Imaging and Patient Safety

X-ray machine (generation and detection); Computed Tomography (CT); Magnetic Resonance Imaging (MRI); Ultrasound imaging; Patient monitoring systems; Electrical safety: Micro-shock, Macro-shock, Grounding, Isolation.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Origin of bio-potentials	Bio-potentials are generated by ion movement across cell membranes.
Physiological systems	The cardiovascular system pumps blood, the nervous system controls functions through electrical impulses, and the respiratory system manages gas exchange.
Bio-electrodes	Electrodes convert ionic current in the body to electronic current in the instrument.
Medical transducers	Transducers convert physiological parameters like blood pressure, body temperature and respiratory flow into electrical signals.
Cardiac therapeutic devices	Pacemakers provide electrical pulses to maintain heart rhythm.
Respiratory and renal support	Ventilators assist or replace spontaneous breathing.
Diagnostic imaging techniques	X-rays use ionizing radiation for bone and tissue imaging.
Hospital safety and monitoring	Patient safety is paramount.

LEARNING RESOURCES

References

Recommended biomedical-engineering references

1. L. Cromwell, F. J. Weibell and E. A. Pfeiffer, Biomedical Instrumentation and Measurements.
2. R. S. Khandpur, Handbook of Biomedical Instrumentation.
3. John G. Webster, Medical Instrumentation: Application and Design.

4. Joseph J. Carr and John M. Brown, Introduction to Biomedical Equipment Technology.

Approved public biomedical-engineering sources

- https://onlinecourses.nptel.ac.in/noc25_bt49/preview
- https://nptel.ac.in/courses/108105101

These sources provide the verified study basis while official Course 5043C detail is unavailable; they do not replace official clinical protocols, authorised device calibration, certified safety standards or authorised hospital plans.